

VARIATION & SELECTION

variation & selection

Variation:

the differences between individuals of the same species

Continuous variation:

the many very small degrees of differences in features between individuals of the same species. (in a range of phenotypes)

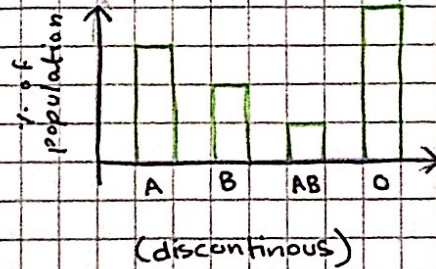
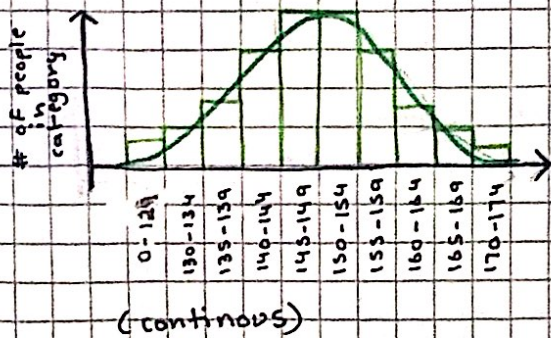
Discontinuous variation:

the distinct differences for a characteristic between individuals of the same species. (in a limited version of phenotypes)

continuous variation = body length, body mass etc.

discontinuous variation = ABO blood groups, seed shape/color etc.

* discontinuous variation is usually caused by genes only while continuous variation is caused by both genes and the environment



Mutation:

the genetic change in which new alleles are formed. Ionising radiation and some chemicals increase the rate of mutation.

Gene mutation:

random change in the base sequence of DNA

* mutation, meiosis, random mating, random fertilisation are sources of genetic variation in populations.

Adaptive Features:

inherited features that help an organism to survive and reproduce in its environment

Hydrophytes

- survive in aquatic environments
- plenty of moisture
- requires a lot of adaptation
- true roots/root hairs are absent
- lack cuticles
- stomata on the upper epidermis

Xerophytes

- survive in dry conditions
- dry
- requires little adaptation
- long roots
- thick cuticles to prevent evaporation
- stomata on the lower epidermis

Natural selection::

- genetic variation within population
 - ↳ many offspring produced
 - ↳ competition for resources — struggle for survival
 - ↳ individuals more adapted to the environment have a greater chance of reproduction than others.
 - ↳ these individuals pass on their genes/alleles to their next generation
- * adaptive features will change due to this — this is evolution
- * over many generations population becomes more suited to their environment.

Selective breeding::

- selection done by humans, of individuals with desirable features
 - ↳ crossing these individuals to produce the next generation
 - ↳ genetics will be carried over to the offspring thus also having desirable characteristics.
- * selective breeding by artificial selection is carried out over many generations to improve crop plants and domesticated animals

→ an example of natural selection is the development of antibiotic resistance by bacteria.

Difference between natural selection and artificial selection :-

<u>natural selection</u>	<u>artificial selection</u>
• occurs naturally • results in development of population with features that are better adapted to their environment • usually takes a long time to occur	• only occurs when humans intervene • results in development of population with features that are useful to humans and not for survival • takes less time